

Formation Inference and Transition Detection from Broadcast Football Video

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Abstract—Football teams are rarely as static as their pre-match formations suggest. In practice, a team’s spatial organization changes constantly, adapting to possession phases, defensive transitions, and the tactical demands of each moment in the match [1]. These formation shifts are typically identified manually and often rely on proprietary tracking data that remains inaccessible to the broader research community [1]. This paper presents a computer-vision pipeline that infers team formations and detects tactical transitions directly from broadcast match video, offering a reproducible alternative to closed tracking infrastructure.

The system detects and tracks players across frames, assigns team identities through a hybrid model-based and jersey-color clustering strategy, and projects player positions onto a normalized pitch coordinate system via homography transformation. Camera motion is compensated using sparse optical flow, and formation labels are inferred by clustering player positions along the depth axis and matching the resulting line structure against standard formation templates using a symmetric Chamfer distance with composite penalty scoring.

The framework was evaluated on 50 publicly available broadcast football video clips across diverse match conditions. Player detection achieved 96.47%, referee detection 91.89%, and ball detection 86.08%. Team assignment accuracy reached 79.25%, and ball possession attribution approximately 77%. Formation transition detection achieved a precision of 67.64%. This work demonstrates that meaningful tactical insight can be extracted from broadcast footage alone, without access to specialized tracking equipment or proprietary datasets.

Index Terms—Football analytics, computer vision, formation inference, tactical analysis, multi-object tracking, YOLOv8, ByteTrack, homography transformation, player trajectory analysis, broadcast video analysis.

I. INTRODUCTION

Football is a game of constant adaptation. A team’s spatial organization shifts continuously throughout a match in response to possession, defensive pressure, and the tactical demands of each moment. While coaches typically describe their teams using a single formation label, a 4-3-3, a 4-4-2, or a 5-3-2, this description captures only a rough approximation of how players actually position themselves during live play. As Shaw and Glickman noted, such labels are crude summaries of player configurations that are significantly more fluid and dependent on game state than a three-digit code would suggest [2]. Modern managers frequently adjust their team’s structure mid-match, switching between defensive and offensive shapes,

reacting to scorelines, and adapting to the opposition. Understanding these tactical shifts is one of the central challenges of football analysis.

The importance of formation analysis extends well beyond academic interest. Tactical decisions about which formation to deploy, when to change it, and how to exploit an opponent’s structure directly influence match outcomes. Bauer et al. showed that formation matchups directly affect expected goals, with some defensive structures performing significantly better against certain attacking formations [1]. For coaching staff at professional clubs, this kind of insight is increasingly built into match preparation. However, accessing the data required to perform such analysis has historically been a significant barrier. Comprehensive quantitative formation analysis has long depended on proprietary player-tracking systems installed in professional stadiums, recording positions at high frequency with centimeter-level accuracy [1][2]. For student researchers, independent analysts, and clubs outside the top professional leagues, this infrastructure is simply out of reach.

A parallel body of work has explored the use of computer vision to extract player information directly from match footage. Object detection and multi-object tracking methods have matured significantly in recent years, enabling reliable detection of players, referees, and the ball from video frames. However, most existing formation analysis research operates exclusively on clean, pre-processed tracking data. The problem of inferring formations directly from raw broadcast video, without access to any proprietary infrastructure, remains comparatively underexplored.

This paper addresses that gap by presenting a pipeline that extracts player trajectories from ordinary broadcast footage and uses them to infer team formations and detect tactical transitions, with no proprietary data required. The work is guided by three research questions:

- 1) Can player trajectories be reliably extracted from broadcast match video using computer vision techniques?
- 2) Can team formations be inferred from the resulting spatial distributions using clustering and spatial modeling methods?
- 3) Can significant formation changes over time be detected from the inferred formation sequences?

The remainder of this paper is organized as follows. Section 2 reviews related work in formation analysis and player tracking. Section 3 describes the methodology of the pipeline in detail. Section 4 presents experimental results and evaluation. Section 5 discusses findings, limitations, and directions for future work.

II. LITERATURE REVIEW

Understanding how football teams organize themselves spatially has been a research interest for well over a decade, but the methods used to study it have evolved significantly, and not always in directions that make the work accessible. This review covers the three bodies of work most relevant to this project: formation analysis using tracking data, player detection and tracking from broadcast video, and homography-based pitch mapping for moving cameras.

2.1 Formation Analysis in Football

The earliest systematic attempts to quantify football formations treated them as static structures. A team would be assigned a single label, 4-4-2, 4-3-3, and so on, based on the average positions of players across a match. Shaw and Glickman were among the first to challenge this approach in a principled way, arguing that such labels are crude summaries of player configurations that are significantly more fluid and dependent on game state than a three-digit code would suggest [2]. Their work introduced a data-driven framework for measuring formations dynamically, using tracking data from 180 professional matches sampled at 25Hz. By computing vectors between neighboring players and applying agglomerative hierarchical clustering with the Wasserstein distance metric, they identified distinct offensive and defensive formation templates and detected when teams made significant tactical changes during a match [2]. This methodology established the technical foundation that much of the subsequent formation analysis literature, including this project, builds upon.

Bauer et al. extended this line of work by introducing the concept of phase-contextualized formation analysis [1]. Rather than measuring formations across the full match, they first decomposed each game into tactical phases, including build-up, mid-block, high-block, low-block, and attacking play, using a convolutional neural network trained on expert-annotated Bundesliga data. Formations were then measured separately within each phase, revealing that a team's spatial structure varies substantially depending on whether they are in possession, pressing, or defending deep [1]. Applied to seven seasons of German Bundesliga data comprising over 2,000 matches, their approach demonstrated that formation pairings have measurable effects on expected goals, and that coaches exhibit consistent tactical fingerprints across multiple seasons. This work represents the most comprehensive formation analysis published to date.

Narizuka and Yamazaki contributed a complementary approach based on Delaunay triangulation, defining a team's formation at any moment as the adjacency matrix of a Delaunay network constructed from player positions [6]. This allows formation similarity to be quantified mathematically

and enables hierarchical clustering across multiple games, something earlier methods struggled with. Applied to J-League tracking data, their algorithm identified five average formation types and further subdivided each into more specific patterns reflecting positional exchanges between players [6]. The concept of role representation they adopt, labeling players by positional function rather than jersey number, is directly relevant to how formation lines are assigned in this project.

What these three studies share, and what limits their broader applicability, is their dependence on professional-grade tracking infrastructure. Shaw and Glickman used data sampled at 25Hz from a single elite league season [2]; Bauer et al. relied on seven seasons of Bundesliga positional data collected by Chyronhego's TRACAB optical tracking system [1]; Narizuka and Yamazaki worked with centimeter-accurate player coordinates provided by DataStadium Inc. under a proprietary agreement [6]. None of these datasets are publicly available, and none of these methods were designed to operate on raw video. This project addresses that constraint directly by operating entirely on broadcast footage.

2.2 Player Detection and Tracking from Broadcast Video

Extracting player positions from broadcast footage is a substantially harder problem than working with clean tracking data, and the computer vision community has been grappling with it for nearly two decades. Beetz et al. presented one of the earliest systems capable of tracking players from live TV broadcast footage, demonstrating player detection rates exceeding 90% on World Cup 2006 data [3]. Their system, ASPOGAMO, estimated camera parameters frame by frame using field line correspondences and optical flow, detected players through color-based segmentation, and tracked them using a Multiple Hypothesis Tracker. The challenges they documented, including camera motion, zoom changes, motion blur, partial occlusions, varying lighting, and noisy compressed video, remain the defining difficulties of broadcast tracking to this day [3].

Mackowiak et al. approached the same problem from a different angle, applying Histogram of Oriented Gradients descriptors combined with Support Vector Machine classification to detect and segment players in broadcast football video [4]. Their work explicitly acknowledged the conditions that make broadcast video uniquely difficult: cameras are not fixed, shots change dynamically according to a broadcast director's decisions, multiple players move simultaneously in different directions, and the background changes rapidly [4]. These are not edge cases, they are the standard operating conditions of any publicly available football footage.

Both systems represent the pre-deep-learning era of broadcast tracking. They relied on hand-crafted features, color models, and classical probabilistic tracking, and while they achieved reasonable detection rates under controlled conditions, they required semi-automatic initialization steps and struggled with robustness across different match conditions. The arrival of deep learning-based object detection, and in particular architectures like YOLOv8 combined with multi-object tracking algorithms like ByteTrack, has dramatically improved

detection accuracy and tracking stability. This project makes use of these modern tools, while the foundational challenges documented by Beetz et al. [3] and Mackowiak et al. [4] remain relevant benchmarks against which progress can be measured.

2.3 Homography Estimation for Moving Cameras

Mapping player positions from image coordinates to a standardized pitch plane requires estimating the homography between the camera view and the field model. This is straightforward when cameras are fixed and field lines are clearly visible, but broadcast footage provides neither guarantee. Gomes et al. identified this as an open problem and proposed using motion information, extracted from optical flow or MPEG-encoded motion vectors, to improve homography estimation in frames where pitch keypoints are sparse [5]. Their work directly targets the scenario that broadcast video presents: a moving camera panning across the field, with large regions of the midfield offering few reliable reference points for traditional keypoint-based calibration [5].

Their results showed that integrating motion data into homography calculations produced more consistent estimates across frames with limited keypoints, outperforming state-of-the-art frame-by-frame methods in those specific conditions [5]. They also demonstrated that MPEG motion vectors can achieve accuracy comparable to optical flow at a fraction of the computational cost, a practical consideration for any pipeline intended to run on standard hardware [5]. The homography strategy adopted in this project, using optical flow for camera motion compensation followed by perspective transformation to a canonical pitch frame, reflects the core insight of this work.

2.4 Positioning This Work

Taken together, the existing literature reveals a clear structural gap. Formation analysis has produced sophisticated and well-validated methods, but all presuppose access to clean, high-frequency tracking data unavailable outside professional clubs and their data partners. The computer vision literature has shown that player tracking from broadcast video is feasible, yet existing broadcast systems stop at detection and position extraction without connecting to formation inference. No published work attempts to close both gaps in a single reproducible pipeline operating end-to-end from ordinary broadcast footage to formation labels and transition detection.

This project occupies that space, combining modern detection and tracking with pitch normalization and spatial clustering for formation inference. The formation measurement approach draws on the methodology of Shaw and Glickman [2] and Bauer et al. [1], the computer vision components build on the broadcast tracking lineage of Beetz et al. [3] and Mackowiak et al. [4], and the pitch mapping strategy is grounded in the motion-aware homography work of Gomes et al. [5]. What is new is the integration of all these components into a single accessible system that requires no proprietary data.

III. METHODOLOGY

3.1 Overview

This project develops a computer-vision pipeline for tactical football analysis from broadcast video. The core objective is to automatically detect and track match entities, such as players, referees, and the ball, assign team identities, estimate player movement, and infer team formations over time, including detecting when those formations change. Rather than relying on any proprietary tracking infrastructure, the entire pipeline operates on a single broadcast video file and requires no external positional data.

The methodology brings together four distinct technical domains: deep learning-based object detection, multi-object tracking for temporal identity consistency, geometric correction for camera-induced distortion, and a combination of rule-based and clustering-based formation inference. Each of these components addresses a specific challenge that broadcast video presents, and they are assembled into a sequential pipeline where each stage builds directly on the outputs of the previous one.

3.2 Data Source and Experimental Setup

The system takes a single broadcast match video as its only input and processes it frame by frame in offline mode - that is, the full video is available before processing begins, which allows the pipeline to make use of temporal context across frames. This is an intentional design choice: offline processing permits trajectory smoothing, ball interpolation, and transition confirmation over time windows that would not be possible in a real-time setting.

The pipeline produces two output artifacts. The first is an annotated video containing player IDs, team color assignments, ball possession indicators, speed overlays, team structure links, and the current inferred formation for each team. The second is a CSV log of confirmed formation transitions, recording for each event the team, the previous formation label, the new formation label, the frame at which the candidate transition was first detected, the frame at which it was confirmed, and the stability duration. These structured outputs are designed to support both qualitative inspection and quantitative evaluation.

The system was implemented in Python using the following tools and frameworks: YOLOv8 (via the Ultralytics library) for object detection, ByteTrack for multi-object tracking, OpenCV for video processing, optical flow computation, and homography transformation, scikit-learn for K-Means clustering used in team assignment, and NumPy for all numerical computation. Processing was conducted on publicly available football video clips.

3.3 Pipeline Architecture

The pipeline is sequential and stateful. Each stage enriches a shared data structure - indexed by track identity and frame number - with new attributes. Early stages produce geometric and identity information; later stages produce semantic and tactical labels. The eleven processing stages run in the following order: video loading, player and ball detection and tracking, position extraction from bounding boxes, camera

motion estimation and trajectory correction, perspective transformation into field coordinates, ball interpolation for missing detections, speed and distance estimation, team assignment, ball possession attribution, tactical formation inference, and finally visualization and export. This ordering is not arbitrary - formation inference and speed estimation both depend on corrected, pitch-normalized coordinates, which are only available after the camera compensation and homography stages have run.

Table 1 summarizes the twelve functional components of the pipeline, their responsibilities, and the data they consume and produce. Each component consumes existing track attributes and appends new ones, which allows individual modules to be developed, tested, or replaced independently without rewriting the full pipeline.

3.4 Object Detection and Multi-Object Tracking

Player, referee, and ball detection is handled by a YOLOv8-based detector, which produces per-frame bounding boxes with class labels and confidence scores. YOLOv8 was selected over earlier YOLO variants and alternative detectors because it offers strong detection accuracy at practical inference speeds and has been widely validated on sports video. Detection runs in mini-batches to improve throughput.

To maintain consistent player identities across frames, ByteTrack is applied to the detection outputs. ByteTrack associates detections with existing tracks using both high-confidence and low-confidence detections, which makes it more robust to the partial occlusions and brief disappearances that are common in broadcast footage. This transforms frame-local bounding boxes into persistent tracks that each carry a unique identity throughout the video. Goalkeeper detections are merged into the player tracking stream for continuity, and the ball is maintained as a separate track entity. Track dictionaries are aligned to video length to avoid downstream indexing errors.

This combination of YOLOv8 and ByteTrack represents the current standard in sports multi-object tracking and directly addresses the limitations of the pre-deep-learning broadcast tracking systems described in the literature review - systems like ASPOGAMO that required semi-automatic initialization and hand-crafted color features [3].

3.5 Geometric Position Modeling

Once tracks are established, each object is reduced from a bounding box to a single representative point. For players and referees, this is the bottom-center of the bounding box - the foot-point approximation - which represents the player's ground contact location. For the ball, the center of the bounding box is used. These point representations are more appropriate for distance computation and tactical geometry than full boxes, since they correspond to actual locations on the pitch surface rather than image-space extents.

3.6 Camera Motion Compensation

Broadcast cameras pan, tilt, and occasionally zoom to follow the play, which means that apparent player movement in image coordinates includes a global camera-induced component that does not reflect actual movement on the pitch. Without correcting for this, speed estimates would be systematically

inflated and formation geometry would drift with the camera rather than reflecting true player positions.

To address this, the pipeline estimates frame-to-frame camera displacement using sparse optical flow. For each consecutive frame pair, grayscale versions of the frames are extracted and stable visual features are identified in predefined regions of the frame - areas less likely to contain moving players. These features are tracked using the pyramidal Lucas-Kanade optical flow algorithm. The dominant displacement vector across tracked features is taken as the camera movement estimate for that frame, and this vector is subtracted from all object positions to produce camera-adjusted trajectories. This approach is consistent with the motion compensation strategy validated by Gomes et al. for broadcast moving-camera scenarios [5].

3.7 Perspective Transformation to Field Coordinates

Camera-adjusted image coordinates are still in pixel space, which is not metrically meaningful - distances and spatial relationships depend on camera angle, zoom level, and distance from the pitch. To convert player positions into a tactically interpretable coordinate system, the pipeline applies a homography-based perspective transformation that maps image points to a canonical pitch frame of standard dimensions (105×68 metres, consistent with FIFA regulations).

The transformation is computed by detecting the field region using a dedicated field detector, identifying the largest detected field bounding box, constructing source vertices from its corners, and mapping these to the corresponding target vertices in the canonical pitch coordinate system. All camera-adjusted player and ball positions are then passed through this transformation to produce pitch coordinates that are directly comparable across frames and geometrically meaningful for formation analysis. This approach reflects the foundational role of homography in sports video analysis documented by Gomes et al. [5] and is in the same spirit as the field model matching used by Beetz et al. in their earlier broadcast tracking system [3].

3.8 Ball Trajectory Completion

Ball detections are frequently missing due to occlusion, motion blur, and the ball's small size relative to the frame. To ensure continuous ball position estimates for possession attribution, the pipeline performs temporal interpolation over the per-frame ball position table. Missing values are filled by linear interpolation between the nearest valid detections, and initial gaps at the start of the video are backfilled. This guarantees a complete ball trajectory for all downstream modules.

3.9 Team Identification

Assigning players to teams from broadcast video is a non-trivial problem because team identity must be inferred visually without access to any metadata. The pipeline uses a hybrid strategy that prioritises model-based classification but falls back to unsupervised color clustering when the model output is unavailable or uncertain.

When a team-classification model output exists and its bounding box overlaps a tracked player with sufficient Inter-

TABLE I
PIPELINE COMPONENT SUMMARY

Component	Responsibility	Main Input	Main Output
Video Loader	Decode input video into frame array	Raw video file	Ordered frame list
Detector + Tracker	Detect entities and preserve IDs	Frame list	Player/referee/ball tracks with bounding boxes
Camera Motion Module	Estimate camera displacement	Frames + positions	Camera-adjusted positions
View Transformer	Map image points to pitch plane	Adjusted positions + field detection	Transformed field coordinates
Ball Interpolator	Fill missing ball detections	Sparse ball track	Continuous ball track
Speed/Distance Module	Compute motion metrics	Transformed trajectories	Speed + cumulative distance
Team Assigner	Label team identity	Player crops + optional model detections	Team ID + team color
Ball Assigner	Assign possession to player/team	Ball position + player positions	Possession flags + possession timeline
Team Structure Drawer	Visualize team connectivity	Team-labeled player positions	Structural overlay
Formation Engine	Infer and smooth formations	Team transformed coordinates	Team formations + transition events
Renderer + Export	Draw all overlays and save artifacts	Enriched tracks + frames	Output video + transition CSV

section over Union, the model label is trusted directly. When model assignment is unavailable or falls below the confidence threshold, the system crops the player region, focuses on the top half of the crop to isolate the shirt area, applies K-Means clustering to separate foreground jersey pixels from background, and then runs a second K-Means pass over all player color samples to split them into two team groups. To avoid initialization failure on frames with very few visible players, the system scans forward until it finds a frame with at least two valid player crops before fitting the fallback clustering model. This hybrid approach is more robust than relying on either method alone - pure color clustering can fail when jersey colors are similar or lighting is inconsistent, while pure model-based assignment can fail on unusual kit combinations. The combination substantially reduces both failure modes.

3.10 Ball Possession Attribution

Per-frame ball possession is assigned to the player whose foot-point is closest to the current ball position estimate, subject to a maximum distance threshold. If no player falls within the threshold, the previous team’s possession state is carried forward. This produces a continuous possession timeline used for displaying live possession percentages in the annotated output.

3.11 Physical Metrics: Speed and Distance

Player speed is estimated from pitch-coordinate displacements over a sliding frame window. The instantaneous speed in metres per second is computed as:

$$v_{m/s} = \frac{d}{\Delta t}$$

where d is the Euclidean displacement over the window in pitch coordinates and Δt is the elapsed time derived from the frame count and the video frame rate. Speed in km/h is obtained by multiplying by 3.6. Cumulative distance is tracked per player identity across the full video and displayed as a contextual physical performance indicator in the annotated output.

3.12 Tactical Formation Inference

Formation inference is the central analytical component of the pipeline and combines spatial clustering with template matching. For each team at each frame, the following steps are applied: the transformed pitch coordinates of all outfield players are gathered, the attacking direction is normalized so that both teams are compared in a shared orientation, goal-keeper candidates not already filtered upstream are removed, player positions along the depth axis of the pitch are clustered into tactical lines, and the resulting line structure is compared against a library of canonical formation templates.

The similarity between an observed player distribution P and a formation template T is calculated using a symmetric Chamfer-like distance:

$$d_{SC}(P, T) = \frac{1}{2} \left(\frac{1}{|P|} \sum_{p \in P} \min_{t \in T} \|p - t\|_2 + \frac{1}{|T|} \sum_{t \in T} \min_{p \in P} \|t - p\|_2 \right)$$

This measures how well the observed player positions match the template in both directions, how far each observed player is from the nearest template position, and how far each template position is from the nearest observed player.

The final candidate score incorporates additional penalty terms beyond the geometric distance:

$$S = d_{SC} + \lambda_1 E_{line} + \lambda_2 E_{count} + \lambda_3 E_{quality} + \lambda_4 E_{transition}$$

where E_{line} penalizes mismatches in the number of players per line, E_{count} penalizes total player count deviations, $E_{quality}$ penalizes low-confidence detections contributing to the estimate, and $E_{transition}$ applies a smoothing penalty that discourages formation changes unless the evidence is strong.

The confidence associated with a formation assignment is derived from the composite score as:

$$c = \text{clip} \left(1 - \frac{S}{0.35}, 0, 1 \right)$$

This scoring design was chosen over a pure geometric matching approach because real broadcast footage frequently produces noisy, incomplete player sets, frames where some players are occluded, or temporarily lost by the tracker. The penalty terms make the matching robust to these conditions without requiring perfect detection in every frame.

3.13 Temporal Smoothing and Formation Transition Confirmation Raw frame-level formation predictions are inherently noisy. A single frame where a player drifts out of position or a detection is missed can produce a spurious formation label that does not reflect any genuine tactical change. To filter out this noise, the pipeline applies confidence-weighted historical smoothing with recency weighting and player-count weighting, so that frames with more visible players and higher matching confidence contribute more strongly to the running formation estimate.

Recency weighting is applied using a linearly increasing function over the history window:

$$w_r(i) = 0.65 + 0.35 \times \frac{i+1}{50}$$

where i is the frame index within the history window, with $i = 0$ being the oldest frame and $i = 49$ the most recent. This ensures that recent frames contribute up to 1.52 times more than the oldest, while old frames are not discarded entirely.

Player-count weighting scales the contribution of each frame by the number of visible players, capped at full trust when ten or more players are detected:

$$w_c = \min\left(\frac{n}{10}, 1.0\right)$$

where n is the number of detected outfield players in that frame. Frames where few players are visible, such as during a camera cut or close-up, carry less influence on the running formation estimate.

Formation changes are governed by a transition state machine. A new formation candidate is not confirmed immediately upon appearing. It must persist for a minimum number of consecutive stable frames before it is accepted as a genuine tactical change. Once confirmed, the event is written to the transition CSV log with full metadata. This minimum stability requirement is the key mechanism for distinguishing real formation changes from detection noise, and it directly addresses the temporal instability problem that any frame-by-frame formation classifier faces when operating on broadcast footage.

3.14 Output and Annotation

The final rendering stage overlays the following information onto the output video: player markers and persistent track IDs, team color indicators and possession highlights, ball marker, player speed text, team structure links connecting players within each formation line, and formation labels with transition status for each team. The annotated video and the transition CSV are exported as the two primary deliverables of the pipeline.

IV. RESULTS

4.1 Overview

This section presents the experimental results obtained by running the pipeline on 50 publicly available broadcast football video clips of approximately 30 seconds each, sourced from footage representing a range of matches, camera angles, and lighting conditions. Manual verification was conducted by the author for each clip, providing ground truth labels against which the system outputs were compared. The evaluation covers six dimensions: player and entity detection, team assignment, ball possession attribution, and formation transition detection.

4.2 Detection Performance

The detection stage was evaluated on its ability to identify players, referees, and the ball within broadcast frames. Table II summarizes the detection accuracy across the evaluation set.

TABLE II
DETECTION ACCURACY BY ENTITY CLASS

Entity	Detection Accuracy
Players	96.47%
Referees	91.89%
Ball	86.08%

Player detection achieved the strongest result at 96.47%, consistent with expectations given that players are the largest and most visually consistent objects in broadcast footage. Referee detection at 91.89% was slightly lower, likely due to the smaller number of referees per frame and occasional visual similarity with players in certain kit combinations. Ball detection at 86.08% was the weakest of the three entity classes, which is expected given the ball's small size, frequent occlusion by players, and motion blur during fast play. Missing ball detections were compensated by the trajectory interpolation module, which filled gaps through linear interpolation to ensure a continuous ball position estimate for downstream possession attribution.

4.3 Team Assignment and Possession Attribution

Team identity assignment was evaluated by comparing the system's per-player team labels against manually verified ground truth labels across the 50 evaluation clips. The hybrid assignment strategy achieved an overall team assignment accuracy of 79.25%. In clips where jersey colors were visually distinct and lighting was consistent, the system performed reliably. Clips with similar jersey tones or strong shadow regions produced higher error rates, particularly when the fallback color clustering model was the primary assignment mechanism.

Ball possession attribution achieved an accuracy of approximately 77%. This metric is inherently challenging because possession in broadcast footage is often ambiguous even to a human observer, particularly during contested duels, loose ball situations, and moments immediately following a tackle. The proximity-based attribution method used by the pipeline is a practical approximation that performs well under normal play conditions but is less reliable in high-contact scenarios.

4.4 Formation Transition Detection

Formation transition detection was evaluated by manually reviewing all transitions flagged by the system across the 50 evaluation clips. Of all transitions detected, 67.64% were confirmed as genuine tactical formation changes by manual verification. The remaining 32.36% were false positives, triggered by temporary detection noise, player clustering during set pieces, or brief periods where too few players were visible for reliable formation inference. Table III summarizes all evaluation results.

TABLE III
SUMMARY OF PIPELINE PERFORMANCE ACROSS ALL EVALUATION DIMENSIONS

Evaluation Dimension	Metric	Result
Player detection	Accuracy	96.47%
Referee detection	Accuracy	91.89%
Ball detection	Accuracy	86.08%
Team assignment	Accuracy	79.25%
Ball possession attribution	Accuracy	~77%
Formation transition detection	Precision	67.64%

4.5 Summary

The results reveal a consistent performance gradient across pipeline stages. Detection performance is strongest in the early stages, with player detection reaching 96.47%. Performance decreases progressively through team assignment (79.25%), possession attribution (~77%), and transition detection (67.64%), reflecting the cumulative nature of the pipeline: each stage inherits any errors introduced by the stages before it, and formation inference is the final and most semantically complex task. This gradient is expected and provides a clear signal about where future improvements would have the greatest impact.

V. ANALYSIS AND VALIDATION

5.1 Interpretation of Results

The results address the three research questions directly. Player detection at 96.47% confirms that the YOLOv8 and ByteTrack combination is sufficiently robust for broadcast conditions, answering RQ1 affirmatively. Team assignment at 79.25% and transition precision at 67.64% show that formation inference and transition detection are both feasible from broadcast-derived positions, partially answering RQ2 and RQ3. The performance gradient across pipeline stages - strongest at detection, weakest at transition inference - reflects the cumulative nature of error propagation in a sequential system, where each stage inherits uncertainty from the one before it.

5.2 Comparison with Existing Work

Direct quantitative comparison with formation analysis literature is not meaningful, as all existing methods operate on clean 25Hz tracking data rather than broadcast video [1], [2], [6]. A more appropriate comparison is with broadcast tracking systems. Beetz et al. reported detection rates above 90% using hand-crafted methods requiring manual initialization [3]; this pipeline achieves 96.47% without any manual intervention,

reflecting the advancement that deep learning has enabled. The team assignment accuracy of 79.25% reflects the same jersey-color ambiguity challenges documented by Mackowiak et al. [4], confirming that this remains a difficult problem even with modern tools. The transition precision of 67.64% has no direct equivalent in the literature, as no published work performs formation transition detection from broadcast video alone.

5.3 Validation and Limitations

Validation relied on manual annotation by the author across 50 broadcast clips, consistent with approaches used in the formation analysis literature [1], [2]. The use of 30-second clips is a practical limitation - short windows are sufficient to observe stable formations but do not capture full tactical evolution across a match. Inter-annotator agreement was not measured, which is a known weakness given that formation labeling is inherently interpretive.

The most impactful technical limitation is the homography approximation. Mapping detected field bounding box corners to a canonical pitch frame works well in most frames but degrades when the visible field region is partial, particularly in midfield areas where pitch keypoints are sparse [5]. The second key limitation is the absence of set-piece detection, which is the primary driver of false positive transitions, accounting for the majority of the 32.36% incorrect transition flags.

5.4 Did the Project Meet Its Objectives?

Yes, with qualifications. All three research questions were answered: player trajectories were reliably extracted, formations were shown to be inferable, and transitions were detectable. The pipeline is fully open-source, requires only broadcast video as input, and delivers interpretable outputs. This fulfills the core contribution of making tactical football analysis accessible without proprietary infrastructure.

VI. CONCLUSION

This project demonstrated that tactical football formation analysis can be performed directly from broadcast video without proprietary tracking infrastructure. A modular pipeline was developed that detects players, assigns team identities, maps positions to a canonical pitch coordinate system, and infers formations with transition detection, operating entirely on publicly available footage.

Player detection reached 96.47%, confirming that the deep learning foundation is robust for broadcast conditions. Performance decreased progressively through semantic stages, with transition precision at 67.64%, reflecting the cumulative difficulty of high-level tactical inference from single-camera data. All three research questions were answered, and the pipeline delivers a fully reproducible, open-source workflow that makes tactical analysis accessible without professional tracking systems.

The primary limitations are the homography approximation in midfield regions, the absence of set-piece detection, and single-annotator evaluation. Future work should focus on shot-type classification to reduce false positive transitions, motion-

aware homography for improved pitch mapping, and extending the pipeline to full match processing. The broader implication is that meaningful tactical insight is achievable from broadcast footage alone, at a fraction of the cost and complexity of professional tracking infrastructure.

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